

Brazil’s soy land footprint is shifting toward China and toward the agricultural frontier

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Abstract

TODO — Data-only working document. The full abstract lives in the main manuscript. Placeholder so the file compiles standalone.

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1 Data

The model is built entirely from publicly and freely available data. Four groups of inputs are combined: (i) Brazilian subnational statistics on soy production, processing, domestic use and trade; (ii) national commodity balances and feed parameters; (iii) administrative and environmental geographies; and (iv) the global input–output backends and the independent validation benchmark. Unless otherwise stated, the series are annual, cover the period 2001–2022 and are broken down by Brazil’s approximately 5,570 municipalities (*municípios*). Where a source exists for only a single reference year, that year is stated and the layer is held constant. Table 1 summarises every input, and all access links are collected in Appendix A. A downloader for the machine-retrievable sources is included with the code.

1.1 Brazilian subnational statistics

Production and population (IBGE): Municipal soybean production and the planted and harvested area data come from the SIDRA system (Table 1612), obtained via its public API. The resident population (Table 6579; 2022 Census) serves as a proxy for household soy oil consumption.

Livestock (IBGE): Municipal herds for ten animal categories (Table 3939) and milked cow counts (Table 94) provide the feed demand base. Feedlot cattle, last reported in the 2006 agricultural census (the 2017 census dropped the confinement question), are carried forward from the 2006 municipal pattern using the national ABIEC/Athenagro confinement series (see Methods).

Processing and biodiesel capacity (ABIOVE, ANP): Soy crushing and refining capacity comes from two manually downloaded ABIOVE products: the consolidated state-level installed capacities and the annual plant roster surveys (*Pesquisa de Capacidade*), whose changing sheet layouts are handled by a per-year dispatcher in the code. Refining and bottling capacities come from the refining sheets of the same products. Years for which surveys are missing from the ABIOVE archive (2000–2002, 2016–2017 and 2021) are replaced with the previous available survey. Biodiesel plant capacity and the regional soy feedstock share are taken from the ANP Statistical Yearbook.

Household consumption and storage (IBGE): State-level per capita soy oil acquisition from the Household Budget Survey (POF 2017–18, the most recent) is used to distribute national food oil use and is held constant across years. Grain storage capacity, which allocates stock changes, is aggregated from the georeferenced warehouse registry underlying the IBGE Stocks Survey (2014 snapshot, held constant).

Municipal trade (COMEX/MDIC): Municipality-level exports and imports are sourced from the COMEX Stat customs database and filtered by the HS4 soy codes 1201 (beans), 1507 (oil) and 2304 (cake).

1.2 National balances and feed parameters (FAO)

Commodity balances: The national supply-use balances against which all municipal totals are reconciled are the FAO Food Balance Sheets for Brazil, items 2555 (soybeans), 2571 (soybean oil) and 2590 (soybean cake), taken from the FAOSTAT bulk download.

Bilateral trade: Re-export disentangling draws on the FAOSTAT Detailed Trade Matrix (items 236, 237 and 238 for beans, oil and cake).

Livestock systems and feed rates: Herds are categorised by production system using the Gridded Livestock of the World rasters (GLW3, reference year 2010) [1] and the GLEAM ruminant system map. They are then converted to soy feed demand using GLEAM per-animal intake rates [2]. These are static layers: the resulting system shares are held constant while herd sizes remain annual (see Methods).

1.3 Geographies

Boundaries: Municipal polygons are IBGE administrative meshes retrieved with the *geobr* package [3]; state outlines for mapping use GADM v3.6.

Biomes and frontier: The footprint origin is attributed to biomes using the IBGE 1:250,000 biome layer. The agricultural frontier is the Amazon biome plus MATOPIBA, operationalised as the Cerrado biome within Maranhão, Tocantins, Piauí and Bahia, closely matching the official Embrapa delineation [4].

Grid refinement (optional): A 30 m visual refinement uses MapBiomass annual soy tiles [5], exported via Google Earth Engine. This refinement is illustrative only and does not affect the load-bearing capacity of the results.

Transport network: The reported model routes flows based on straight-line distances between municipal centres (see Methods). Four multimodal layers — OpenStreetMap roads, DNIT waterways, ANTT rail lines and cargo, and ANTAQ ports and port cargo — support an experimental cost-based transport variant stored in the repository and used for descriptive analyses. These layers do not affect the reported results.

1.4 Global models and benchmark

FABIO. Subnational flows are incorporated into the Food and Agriculture Biomass Input–Output model [6]. From 2010 onwards the version 2 build is used (2010–2023; 22,263 processes, 181 regions $\times$ 123 items, with 123 environmental extensions). As the v2 build did not exist before 2010, the years 2001–2009 use the corresponding inputs of the v1.1 build: bilateral trade details, land use extensions, feedstock optimisation results and v1 balanced trade and commodity balance sheets. Version bridging and the quantified v1.1/v2 seam are described in the ‘Methods’ section.

EXIOBASE. The non-soy background economy is EXIOBASE 3 [7] in the product-by-product transformation (2000–2022; 49 regions $\times$ 200 products), hybridised with FABIO for the non-food side of the footprint.

Trase (validation only). Modelled origin-to-importer flows are validated against the Trase Brazil soy composite v2.6.1 (2004–2022), which links producing municipalities to importing countries from per-shipment customs and logistics records [8, 9]. No Trase data enter the model itself.

1.5 Coverage and summary

The Footprint series is reported for the period 2001–2022. FABIO v2 covers the period from 2010 onwards, while the FABIO v1.1 build provides the backend for the period from 2001 to 2009. The year 2000 is excluded pending the resolution of a demand-side allocation issue in that year’s build (see Methods). The Trase benchmark supports validation for 2004–2022.

1.6 Data availability

All input data are publicly and freely available from the providers listed in Appendix A. They are not redistributed with the code because of differing licences; a download script covers the machine-retrievable sources, and the datasets requiring manual retrieval (ABIOVE, ANP, POF, MapBiomass tiles) are documented step by step in the repository. The intermediate and final model outputs underlying the figures will be deposited on Zenodo upon publication.

Table 1 Input datasets, by role. All are publicly and freely available; access links are collected in Appendix A. “Munis” = Brazilian municipalities; “exp.” = experimental variant only.

Dataset	Provider	Role	Coverage
Soy production, area (t.1612)	IBGE (SIDRA)	production	munis, annual
Population (t.6579/9514)	IBGE (SIDRA)	food-use proxy	munis, annual
Herds (t.3939), milk cows (t.94)	IBGE (SIDRA)	feed demand	munis, annual
Feedlot cattle (t.919)	IBGE (Census)	feed demand	munis, 2006 pattern
Crush/refining capacity	ABIOVE	processing	munis, annual
Biodiesel capacity	ANP	other (biodiesel) use	munis, annual
Soy-oil acquisition (POF)	IBGE	food-oil allocation	states, 2017–18
Grain storage (warehouse registry)	IBGE	stock allocation	munis, 2014 static
Commodity balance (FBS)	FAO (FAO-STAT)	national reconciliation	Brazil, annual
Gridded livestock (GLW3)	FAO (Data-verse)	system split	10 km, 2010
Production systems, feed (GLEAM)	FAO	feed rates	10 km, static
Municipal trade	COMEX/MDIC	exports/imports	munis, annual
Road network	OSM (Geofabrik)	transport (exp.)	national
Waterways	DNIT	transport (exp.)	national
Rail lines, stations, cargo	ANTT	transport (exp.)	national, 2006–21
Ports, port cargo	ANTAQ	transport (exp.)	national, 2017–22
Municipal boundaries	IBGE (geobr)	geometry	munis
State boundaries	GADM v3.6	mapping	states
Biomes 1:250k	IBGE	origin attribution	national
Soy land tiles	MapBiomass (GEE)	grid refinement	30 m, annual
FABIO v2 (MRIO)	fineprint	supply-chain nesting, 2010–	global, 2010–23
FABIO v1.1 (MRIO)	fineprint	supply-chain nesting, pre-2010	global, 1986–2013
EXIOBASE 3 pxp	EXIOBASE	non-soy background	global, 2000–22
Detailed trade matrix	FAO (FAO-STAT)	re-exports/trade	Brazil, annual
Trase composite v2.6.1	Trase	validation benchmark	munis, 2004–22

1.7 Code availability

The complete modelling pipeline (data preparation, transport allocation, FABIO/EXIOBASE nesting, footprint accounting, validation and figures) is written in R and Python and released under the GNU GPL v3 at <repository URL --- TODO>. The results reported here have no proprietary software dependency. The experimental multimodal-transport module included in the repository (not used for the reported results) additionally requires a GAMS licence.

A Data access links

All URLs were last verified in July 2026. Brazilian government portals (IBGE, ComexStat) may reject automated requests but resolve in a browser. A programmatic downloader covering the machine-retrievable sources ships with the code (`code/download_data.py`).

A.1 Brazilian subnational statistics

- **Soy production, planted and harvested area** (IBGE SIDRA, table 1612): <https://sidra.ibge.gov.br/tabela/1612>; API pattern <https://apisidra.ibge.gov.br/values/t/1612/n6/all/v/109,216,214/p/{year}/c81/2713>.
- **Population** (IBGE SIDRA, table 6579; 2022 Census table 9514): same API endpoint pattern with the respective table number.
- **Livestock herds** (IBGE SIDRA, table 3939), **milked cows** (table 94), **feedlot cattle** (Census tables 919/6911): same API endpoint pattern.
- **Grain storage** (IBGE Stocks Survey warehouse registry): <https://sidra.ibge.gov.br/tabela/278>.
- **Crushing and refining capacity** (ABIOVE, Capacidade Instalada; manual download): <https://abiove.org.br/estatisticas/>.
- **Biodiesel capacity** (ANP Statistical Yearbook, table 2.6; manual download): <https://www.gov.br/anp/pt-br/centrais-de-conteudo/publicacoes/anuario-estatistico/>.
- **Household soy-oil acquisition** (IBGE POF 2017–2018): <https://www.ibge.gov.br/estatisticas/sociais/educacao/9050-pesquisa-de-orcamentos-familiares.html>.
- **Municipal foreign trade** (COMEX Stat / MDIC): per-year files <https://balanca.economia.gov.br/balanca/bd/comexstat-bd/mun/> (EXP_{year}_MUN.csv, IMP_{year}_MUN.csv); lookup tables <https://balanca.economia.gov.br/balanca/bd/tabelas/> (PAIS.csv, UF_MUN.csv). Directory listings are blocked; the files themselves download normally.

A.2 National balances and feed parameters (FAO)

- **Commodity balance sheets** (FAOSTAT FBS): <https://www.fao.org/faostat/en/#data/FBS>; bulk file [https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_All_Data_\(Normalized\).zip](https://bulks-faostat.fao.org/production/FoodBalanceSheets_E_All_Data_(Normalized).zip).
- **Detailed trade matrix**: bulk file [https://bulks-faostat.fao.org/production/Trade-DetailedTradeMatrix_E_All_Data_\(Normalized\).zip](https://bulks-faostat.fao.org/production/Trade-DetailedTradeMatrix_E_All_Data_(Normalized).zip).

- **Gridded livestock (GLW3)**: <https://dataverse.harvard.edu/dataverse/glw>.
- **GLEAM production systems and feed rates**: <https://www.fao.org/gleam/resources/en/>.

A.3 Transport network layers

- **Roads** (OpenStreetMap, Brazil extract): <https://download.geofabrik.de/south-america/brazil.html>.
- **Waterways (DNIT)**: <https://www.gov.br/dnit/pt-br>.
- **Rail lines, stations and cargo (ANTT)**: <https://dados.antt.gov.br/dataset/transporte-ferroviario-de-cargas-e-passageiros>.
- **Ports and waterway cargo (ANTAQ)**: <https://web.antaq.gov.br/anuario/> (temporarily unreachable at the July 2026 check; re-verify before use).

A.4 Geographies

- **Municipal boundaries** (IBGE, via the `geobr` package): <https://github.com/ipeaGIT/geobr>; <https://cran.r-project.org/package=geobr>; localities API <https://servicodados.ibge.gov.br/api/v1/localidades/municipios>.
- **State boundaries** (GADM v3.6): <https://gadm.org/>.
- **Biomes 1:250,000** (IBGE): <https://www.ibge.gov.br/geociencias/informacoes-ambientais/estudos-ambientais/15842-biomas.html>.
- **Soy land-use tiles** (MapBiomas, via Google Earth Engine): <https://mapbiomas.org/>; export toolkit <https://github.com/mapbiomas-brazil/user-toolkit>.

A.5 Global models and benchmark

- **FABIO** (v2 build and v1.1 release): <https://github.com/fineprint-global/fabio>; v1.1 archive DOI 10.5281/zenodo.2577067.
- **EXIOBASE 3** (pxp): <https://www.exiobase.eu/>; data DOI 10.5281/zenodo.3583070.
- **Trase Brazil soy composite v2.6.1**: <https://trase.earth/>.

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